BHASKAR S

ML ENGINEER

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PROJECTS

- **MicroGPT Reasoning Model** with **Chain-of-Thought**: Engineered a lightweight transformer module incorporating **chain-of-thought** reasoning (inspired by **O1**) to break down complex tasks and support multi-step logical reasoning in natural language.
- **Handwritten Math Solver**: Developed a personal project that converts handwritten equations into computed results using a custom **CNN** for character recognition—utilizing **GANs** for **data augmentation** and **VAEs** for **noise reduction**.
- Developed a diverse portfolio of small-scale AI solutions spanning **computer vision, generative modeling, and natural language processing**, each showcasing **rapid prototyping** and a commitment to **innovative problem-solving**.

TECHNICAL SKILLS

- Programming Languages: **Python**, JavaScript, PHP
- **Machine Learning Frameworks: PyTorch, TensorFlow**
- **Deep Learning Architectures**: **GANs, GPTs, VAEs, Transformers, CNNs, NLP**
- Databases: PostgreSQL, MySQL
- Web Technologies: React, Node.js, Express.js, Vue 3, Pinia, TailwindCSS Tools
- & Platforms: Git, Docker, **AWS**, RESTful APIs
- Other: **Computer Vision**, **Data Analysis**.

PROFESSIONAL EXPERIENCE

Full Stack Developer | Shoptree Technologies Pvt Ltd **Jan 2018 - Present**

- Spearheaded development of the company's primary platform, contributing to nearly **45%** of its codebase and **70%** overall.
- Reduced running costs by approximately **50%** by optimising key database queries, overall system workflows.
- Designed and implemented **scalable system architectures** for management and inventory systems that are both simplified and seamlessly integrable with other applications.
- Developed and maintained critical applications, significantly enhancing operational efficiency.
- Collaborated with cross-functional teams to drive **continuous improvement** and **innovation** in the product suite.

ML Engineer | Shoptree Technologies Pvt Ltd **Dec 2021 - Present**

- Designed and deployed automated ML-driven inventory flows that streamlined stock management, reduced manual intervention, and improved operational efficiency across the production platform.
- Developed and integrated an intelligent chatbot with advanced OCR recognition capabilities for real-time document and data processing, enabling seamless automation of inventory-related workflows.
- Built predictive analytics models for inventory analysis to forecast demand, optimise stock levels, and minimise overstock/understock situations in live production environments.
- Engineered customer churn prediction models using machine learning techniques to identify at risk customers and generate actionable insights for retention strategies.
- Deployed all ML solutions directly into the production platform, delivering end-to-end data-drive automation and measurable business impact

ADDITIONAL INFORMATION

- Deeply committed to advancing knowledge in **AI**, with a particular focus on developing **SAFE AGI** and exploring ways to enhance **human consciousness** and understand the **UNIVERSE**.
- Active solving coding challenges to continuously **sharpen algorithmic** and **problem-solving skills** (basically fine-tuning my biological neural nets).
- Proficient in **English**, **Kannada**, and **Telugu**, with basic proficiency in **Hindi**.
- Passionate about community service, regularly volunteering to support local **educational initiatives** and **help children learn**.
- Discover more about my projects, insights, and blog posts on my personal website: bhaskar0210s.github.io/me

SUMMARY

- **Driven** by an enduring **passion** for uncovering the secrets of the universe, I have spent **several years** exploring **AI** and hands-on experience in developing generative AI models, including building **GANs** for synthetic data generation, **Chain-of-thought** reasoning **GPT** model, and conducting experiments with transformer-based **NLP** models.
- Proficient in **Python**, **PyTorch**, **TensorFlow**, and a range of Python libraries, with strong expertise in designing, training, and evaluating advanced AI models through self-driven projects.
- **Extensive** experience as a **Full Stack Developer** using modern web technologies to build **scalable**, robust, and integrable software solutions.
- Dedicated to bridging the gap between cutting-edge **AI research** and practical application, continuously enhancing skills to solve complex challenges.

EDUCATION

BMS College of Engineering **Aug 2013-2015**

CGPA: 8.5

National Pre-University College Jayanagar **May 2011-2013**

671 CET **State** Rank with **95%** in PCME